

PHY209 Engineering Physics II

Chapter 1

Emergence of Quantum Physics

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1.0 The Breakdown of Classical Physics

During the 19th century many of the great advances in physics of the 18th century were consolidated and extended. In addition, the theory of electromagnetism was completed by J.M. Maxwell. Except for a few unexplained effect or anomalies there seemed little more in terms of fundamental physics to be done by the beginning of the 20th century. Yet it is precisely in the year 1900 that quantum theory starts its journey with Planck's formula for blackbody radiation.

Soon there were a host of experimental results, both new ones and earlier ones that again attracted attention. All of these pointed to deficiencies in the physics of the 19th century. In almost all cases these anomalies resulted when Newtonian mechanics and electromagnetic theory, both together known as classical theory, were simultaneously involved. In trying to elucidate these various experimental facts a new theory of physics, call quantum theory, was born.

The experimental findings of some of the phenomena that cannot be explained by the help of the classical theory are the followings: **Black Body Radiation, Stability of Atoms, Photo Electric Effect, Compton Effect, etc.**

In the next few sections, we briefly examine several of these experiments and discuss them with some modern insight. First, we consider blackbody radiation from the pre-quantum or classical point of view. We then turn to a consideration of the stability of the classical Rutherford atom. Although Rutherford had experimentally demonstrated the planet-like structure of the electrons in atoms, his model caused a lot of theoretical problems. An even older effect, dating back to Herz in 1887, the photoelectric effect, provides another example of the complete breakdown of classical physics. After that we discuss the elastic scattering of light off electrons, the Compton Effect, which also clearly demonstrates the corpuscular nature of light. Finally, we consider the wave-particle duality and the de Broglie hypothesis.

1.1 The Blackbody Radiation

The first definite failure of classical physics was found in the study of the nature of distribution of energy in the spectrum of radiation from a blackbody. When a body is heated it emits electromagnetic radiation and when the temperature is lowered energy is absorbed by the body. In case of heat radiation, when the radiant energy absorbed by a body is equal to the heat emitted, the system will be in thermal equilibrium. If a body absorbs all the radiation that fall upon it, it is known as a blackbody and the radiation emitted by such a body is known as blackbody radiation. In practice, no object is a perfect blackbody. For experimental purposes, a cavity made of a highly insulating wall with a small hole, heated to a constant high temperature would serve as a blackbody. When a cavity is heated, radiation of all wavelengths would be emitted from the hole. The radiation leaking out of the hole has been reflected inside the cavity so many times that it has come to a thermal equilibrium with the walls. Thus, it may be treated as an ideal blackbody radiator.

In 1859, Kirchoff formulated two laws concerning the properties of a blackbody.

- A blackbody not only absorbs all the radiation falling on it but also acts as a perfect radiator when heated.
- The radiation given out by a blackbody is independent on the temperature of the cavity and is independent of the nature of the interior material.

The radiant energy may be analyzed by passing it through a prism and breaking it up into radiations of various wavelengths. The energy associated with different wavelengths can be measured. In the study of blackbody radiation, it is a general practice to study the spectral distribution, i.e., how the energy is distributed among the different wavelengths emitted by a blackbody. **The spectral distribution of blackbody radiation was studied by Lummer and Pringsheim in 1897 experimentally at different temperatures.** Some of these results are shown in the Fig. 1.1 given below. Two principal features are apparent from the results prescribed in the figure.

- The wavelength corresponding to a peak shift from higher to lower values as the temperature is raised. The distribution is wider at lower temperatures.
- Energy density per unit wavelength (known as spectral energy density) is more at high temperatures than at low temperatures.

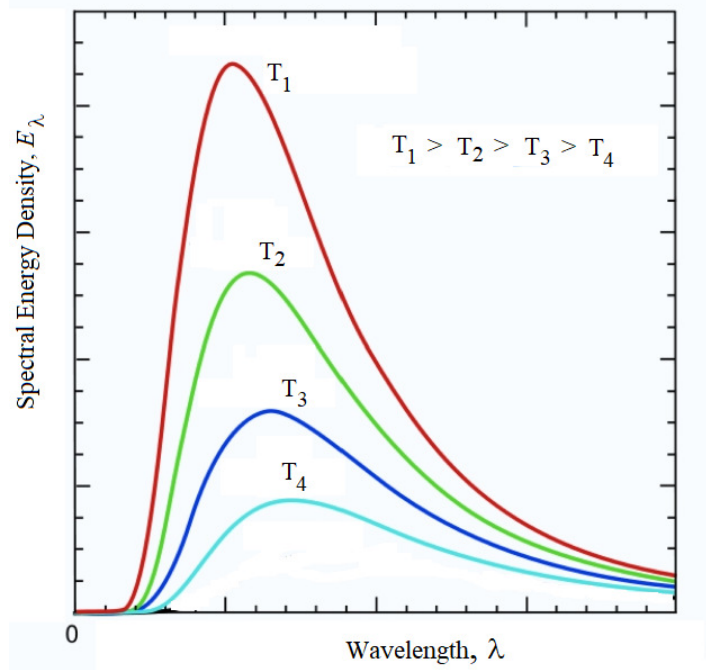


Fig. 1.1: Spectral energy density E_λ of distribution of radiation with respect to the wavelength λ , from a blackbody at different temperatures (Lummer and Pringsheim).

The Classical Theory

Wien Theory

In order to find a theoretical interpretation of the actual distribution, **Wien** in **1893**, derived an expression on the basis of thermodynamic considerations. The amount of energy density $E_\lambda d\lambda$ emitted by a blackbody at a temperature T within the wavelength range λ and $\lambda + d\lambda$, is given by

$$E_\lambda = A_1 \lambda^{-5} e^{-A_2/\lambda T}, \quad (1.1)$$

where E_λ is the energy density of radiation at wavelength λ per unit wavelength, A_1 and A_2 are two constants. From the formula (1.1), we notice that for $\lambda = 0$ or ∞ , $E_\lambda d\lambda = 0$. Thus, no energy is emitted by a wave of zero or infinite wavelength. This is against the actual behavior of a blackbody radiator. Equation (1.1) is found to be satisfactory in the lower range of wavelengths at high temperatures but fails completely while explaining the experimental curve at higher wavelengths as shown in Fig. 1.1. Equation (1.1) is known as the **Wien Law**.

Rayleigh-Jeans Theory

In **1900**, Rayleigh and Jeans applied the principle of equipartition of energy to blackbody radiation. They considered a blackbody radiator to be made up of several oscillators with one possible frequency for each oscillator. According to the principle of equipartition of energy, an oscillator in equilibrium with the source at a temperature T , possesses a mean kinetic energy $k_B T$, where k_B is the Boltzmann constant. The number of oscillators per unit volume, dN , in the wavelength range between λ and $\lambda + d\lambda$ can be calculated based on the classical viewpoint as

$$dN = \frac{8\pi}{\lambda^4} d\lambda. \quad (1.2)$$

Therefore, the energy density in the wavelength range $d\lambda$, $E_\lambda d\lambda$, is simply the product of the number of oscillators per unit volume and the mean energy $k_B T$ of the oscillator:

$$E_\lambda d\lambda = dN(k_B T) = \frac{8\pi}{\lambda^4} k_B T d\lambda,$$

or

$$E_\lambda = \frac{8\pi}{\lambda^4} k_B T. \quad (1.3)$$

As λ decreases, the energy density gradually increases without going through the maximum and ultimately becomes infinite at $\lambda = 0$. Equation (1.3) is known as the **Rayleigh-Jeans Law**. This is not in agreement with the experimental facts. As given in Fig. 1.1, the Rayleigh-Jeans formula agrees fairly well with the experimental results only in the longer wavelength region corresponding to very small frequencies.

Neither of the two equations, Eqs. (1.2) and Eq. (1.3), discussed so far, is consistent with the experimental curve, as shown in Fig. 1.1, over the complete range of wavelengths. Thus, it can be concluded that blackbody radiation cannot be explained using classical mechanics.

The Quantum Theory

Planck's Quantum Hypothesis

The first successful theory for the spectral distribution of blackbody radiation was presented by Planck to the Berlin Physical Society on December 14, 1900. In his revolutionary theory, he discarded the classical concept that an oscillator emits or takes up energy in a continuous fashion. According to this new theory,

- An oscillator absorbs or emits radiation discontinuously, in the form of energy packets, called quanta. Quanta may be considered as behaving like a stream of particles, possessing energy and momentum. The energy of a quantum of radiation is given by

$$E = hf, \quad (1.4)$$

where f is the frequency of radiation and h is a constant (later on it becomes the Planck's constant). In SI units, $h = 6.63 \times 10^{-34}$ Js. Obviously, the magnitude of quantum is not fixed, and it will depend on the frequency of the oscillator.

- The oscillator has definite amounts of energy in discrete levels called energy levels. The energy E_n of the n -th energy level will be an integral multiple of a quantum, i.e.,

$$E_n = nhf, \quad (1.5)$$

where n is an integer, having values 0, 1, 2, 3,

Derivation of the Planck's Formula

Let us consider that a blackbody radiator consists of N simple harmonic oscillators having fundamental frequency f . If these oscillators can take up energy only in multiples of hf , then the allowed energies will be $E_0 = 0, E_1 = hf, E_2 = 2hf, E_3 = 3hf, \dots$, etc. Let $N_0, N_1, N_2, N_3, \dots$, etc. be the number of oscillators with corresponding energies $E_0, E_1, E_2, E_3, \dots$, etc. The number N_j , having energy E_j above the lowest energy E_0 , can be determined using the Boltzmann distribution law

$$N_j = N_0 e^{-E_j/k_B T}. \quad (1.6)$$

Thus, we have

$$\begin{aligned} N_1 &= N_0 e^{-E_1/k_B T} = N_0 e^{-hf/k_B T}, \\ N_2 &= N_0 e^{-E_2/k_B T} = N_0 e^{-2hf/k_B T}, \\ &\dots\dots\dots \\ N_j &= N_0 e^{-E_j/k_B T} = N_0 e^{-jhf/k_B T}, \end{aligned} \quad (1.7)$$

where k_B is the Boltzmann constant.

The total number of oscillators, N , in all the energy states is given by

$$\begin{aligned} N &= N_0 + N_1 + N_2 + \dots + N_j + \dots \\ &= N_0 + N_0 e^{-hf/k_B T} + N_0 e^{-2hf/k_B T} + \dots + N_0 e^{-jhf/k_B T} + \dots\dots \\ &= N_0 (1 + e^{-hf/k_B T} + e^{-2hf/k_B T} + \dots + e^{-jhf/k_B T} + \dots\dots) \\ &= N_0 \sum_{j=0}^{\infty} e^{-jhf/k_B T}. \end{aligned} \quad (1.8)$$

The total energy E of all the oscillators is given by

$$\begin{aligned}
 E &= E_0 N_0 + E_1 N_1 + E_2 N_2 + \cdots + E_j N_j + \cdots \\
 &= 0 \cdot N_0 + hf \cdot N_0 e^{-hf/k_B T} + 2hf \cdot N_0 e^{-2hf/k_B T} + \cdots + jhf \cdot N_0 e^{-jhf/k_B T} + \cdots \\
 &= N_0 hf (0 + e^{-hf/k_B T} + 2e^{-2hf/k_B T} + \cdots + je^{-jhf/k_B T} + \cdots) \\
 &= N_0 hf \sum_{j=0}^{\infty} j e^{-jhf/k_B T}.
 \end{aligned} \tag{1.9}$$

Thus, the average energy, $\langle E \rangle$, of an oscillator with fundamental frequency f is given by

$$\langle E \rangle = \frac{E}{N} = \frac{N_0 hf \sum_{j=0}^{\infty} j e^{-jhf/k_B T}}{N_0 \sum_{j=0}^{\infty} e^{-jhf/k_B T}}. \tag{1.10}$$

If we put, $e^{-hf/k_B T} \equiv x$, then Eq. (1.10) becomes

$$\langle E \rangle = \frac{hf \sum_{j=0}^{\infty} j x^j}{\sum_{j=0}^{\infty} x^j}, \tag{1.11}$$

where the denominator is

$$\sum_{j=0}^{\infty} x^j = (1 + x + x^2 + \cdots) = \frac{1}{1-x} \quad \text{for } x < 1$$

and the numerator is

$$\begin{aligned}
 \sum_{j=0}^{\infty} j x^j &= (0 + x + 2x^2 + 3x^3 + \cdots) = x(1 + 2x + 3x^2) \\
 &= \frac{x}{(1-x)^2} \quad \text{for } x < 1.
 \end{aligned}$$

Using the above expressions, Eq. (1.11) then becomes

$$\langle E \rangle = \frac{hf x (1-x)}{(1-x)^2} = \frac{hf x}{1-x} = \frac{hf e^{-hf/k_B T}}{1 - e^{-hf/k_B T}} = \frac{hf}{e^{hf/k_B T} - 1}. \tag{1.12}$$

Planck's Distribution Equation in terms of Wavelength

Since the relation between the wavelength and frequency is $\lambda = c/f$, Eq. (1.12) can be written as

$$\langle E \rangle = \frac{hc}{\lambda(e^{hc/\lambda k_B T} - 1)}. \tag{1.13}$$

Now, recalling that the number of oscillators per unit volume within the wavelength range between λ and $\lambda + d\lambda$, from Eq. (1.2) is $dN = (8\pi/\lambda^4)d\lambda$. Hence, the energy density ($E_\lambda d\lambda$) in the wavelength range between λ and $\lambda + d\lambda$, is simply the product of the number of oscillators per unit volume and the average energy of oscillator. Thus, the energy density is

$$E_\lambda d\lambda = dN \langle E \rangle = \frac{8\pi}{\lambda^4} d\lambda \frac{hc}{\lambda(e^{hc/\lambda k_B T} - 1)}$$

or

$$E_\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1}. \tag{1.14}$$

This is known as the **Planck's distribution equation** of radiation energy in terms of wavelength. This equation (1.13) agrees with the experimental results of Lummer and Pringheim as is given in Fig. 1.1.

Planck's Distribution Equation in terms of Frequency

Since $\lambda = c/f$, we have $d\lambda = -c df/f^2$ and $E_\lambda d\lambda = -E_f df$, the minus sign represents the inverse relation between wavelength λ and the frequency f . Thus, Eq. (1.14) can be expressed in terms of frequency f as

$$E_f = \frac{8\pi hf^3}{c^3} \frac{1}{e^{hf/k_B T} - 1}. \quad (1.15)$$

Recovery of the Wien Law

The Planck's distribution equation may be used to account for the Wien laws as follows:

when $hc \gg \lambda k_B T$ (i.e., T or λ is small), in that case, $e^{hc/\lambda k_B T} \gg 1$ and we can neglect the 1 in the denominator of Eq. (1.14), so that

$$E_\lambda = \frac{8\pi hc}{\lambda^5} e^{-hc/\lambda k_B T} d\lambda = A_1 \lambda^{-5} e^{-A_2/\lambda T},$$

which is the Wien's equation (Eq. (1.1)), where $A_1 = 8\pi hc$ and $A_2 = hc/k_B$.

Recovery of the Rayleigh-Jean's Law

On the other hand, when $hc \ll \lambda k_B T$ (i.e., T and λ is large), in that case, we have $e^{hc/\lambda k_B T} \approx 1 + hc/(\lambda k_B T)$. Therefore, Eq. (1.14) becomes

$$E_\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{\left(1 + \frac{hc}{\lambda k_B T} - 1\right)} = \frac{8\pi}{\lambda^4} k_B T.$$

That is

$$E_\lambda = \frac{8\pi}{\lambda^4} k_B T.$$

This is the Rayleigh-Jeans Law, Eq. (1.3).

Thus, we find that both the Wien law and the Rayleigh-Jeans law follow from the mathematical Planck's quantum distribution equation. Further, if $h \rightarrow 0$ (for $hc \ll \lambda k_B T$), we have

$$e^{hc/\lambda k_B T} - 1 \approx \frac{hc}{\lambda k_B T}.$$

1.2 The Photoelectric Effect

Just prior to 1900, it is in 1887, Heinrich Hertz performed the experiments which showed that when a beam of visible or ultraviolet light falls on a clean alkali metal surface, electrons are emitted from the surface. This effect is known as the photoelectric effect and occurs in all solids, liquids and gases provided that the radiation of appropriate frequency is used. This was the first instance of light interacting with matter, so it was very mysterious. In 1905 Albert Einstein, a 3rd Class Technical Expert in the Swiss Patent Office and a bureaucrat, published a paper which provided the explanation for the effect - light is made up of small particles.

Surface electrons are bound to metals with a small amount of energy. Some of the incident photons enter the surface, collide with atoms of the metal, and are totally absorbed. They give their energy to an electron, which, if the absorbed energy was great enough, then break free from the atom. The photoelectric effect is the result of collisions between photons and electrons that knock the electrons out of the metal. The energy required to knock out an electron from an atom of the metal surface is known as the work function Φ of the metal. This is the energy binding the electron to the metal. Each photon of light has energy $E = hf$. The electron that gets kicked out of the metal gets its energy from the photon. Some of the energy of a photon is used to break the electron from the metal (that is Φ) and the rest energy goes to have the kinetic energy T_{max} of the emitted photon:

Table 1.1 gives the work functions of several metals.

Table 1.1 Some metals and the corresponding work functions

Metal	Φ (eV)
Sodium	2.28
Zinc	4.31
Cobalt	3.90
Silver	4.73
Platinum	6.35

The typical experimental set up is shown in Fig. 1.2. In a highly evacuated tube, when a beam of ultraviolet light falls on a metal surface, considered as cathode C, electrons are ejected from the surface. These electrons are then pulled to the anode A by a sufficiently high voltage V , thus completing the circuit. The

flow of electron current is then registered by a galvanometer or electrometer G . This current is known as the photoelectric current.

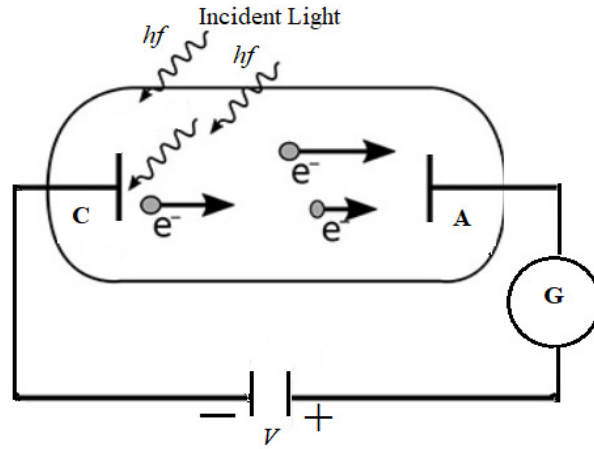


Fig. 1.2: Schematic diagram of apparatus for demonstrating the photoelectric effect.

In the following, we shall discuss two experiments on photoelectric effect.

The First Experiment

When monochromatic radiations of the same frequency but varying intensities are allowed to fall upon the metal surface, we obtain a series of current-potential curves as shown in Fig. 1.3. When V is positive, the current I reaches a limiting value (called saturation current) for each curve. When V is made increasingly negative the current begins to decrease until at a sharply defined value of V , it becomes zero. The gradual decrease in current is since some electrons are ejected from the cathode surface C with sufficient speed to enable them to overcome a retarding potential of a few volts. This limiting retarding potential, at which all electrons are stopped and at which current becomes zero, is called the stopping potential V_0 for that particular surface and frequency.

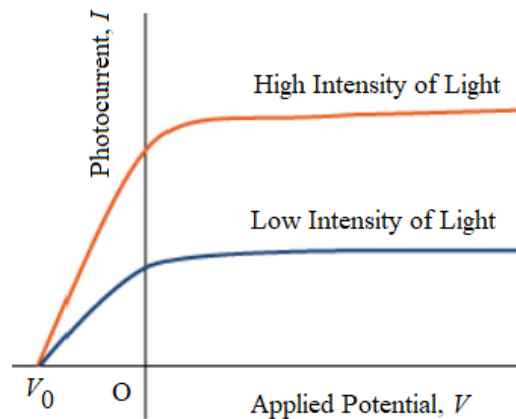


Fig. 1.3: Photoelectric current, I , plotted against the applied potential, V , for two different intensities of illuminating light; V_0 is the stopping potential, which is negative.

Since voltage V_0 is required to stop electrons with maximum speed v_m , this potential is a measure of the maximum kinetic energy of the electrons. By the law of conservation of energy, the stopping potential is thus just equal to the maximum kinetic energy of the electrons, i.e.,

$$\frac{1}{2}mv_m^2 = eV_0. \quad (1.16)$$

It is also apparent from Fig. 1.3 that the stopping potential is independent of the light intensity, but the total photoelectric current is proportional to the intensity.

The Second Experiment

Although the stopping potential is a constant quantity for a given frequency, it varies with frequency. The relation between the frequency f and V is then represented by Fig. 1.4, which indicates a linear relationship between these variables.

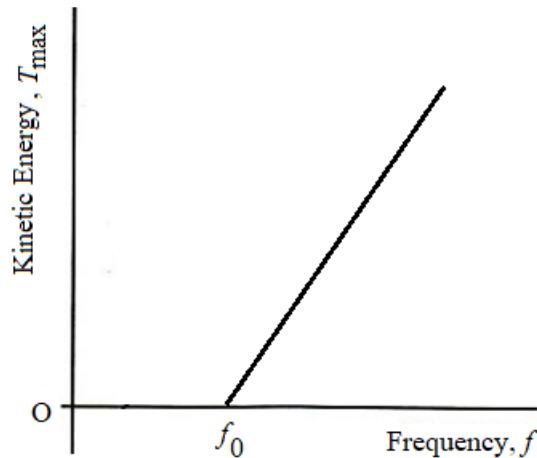


Fig. 1.4: Variation of maximum kinetic energy T_{max} with frequency f of the incident light.

The stopping potential V_0 (or maximum kinetic energy) becomes zero below a sharply defined frequency, indicating that no electrons are emitted. This limiting frequency below which no electrons are emitted, is called the threshold frequency f_0 .

So far, we have reported experimental results using diagrams. We summarize these results as follows:

1. The maximum kinetic energy of the photoelectrons is independent of the intensity of incident radiation but varies directly with its frequency.
2. Only radiation of a certain characteristic minimum frequency can remove electrons from a particular metal surface. This frequency is known as the threshold frequency. Below this frequency, no electrons are emitted, no matter how great the intensity is. The value of the threshold frequency is different for different metals.
3. The total photoelectric current is directly proportional to the number of electrons emitted per unit time and this in turn, is directly proportional to the intensity of the incident radiation.

According to the classical electromagnetic theory, the kinetic energy of the emitted electron should increase with the increase in intensity of the incident radiation. It should be expected that if light were allowed to fall upon a metallic surface for a sufficient length of time, electrons would be emitted regardless of the frequency of the incident light. This is, however, not observed in the experiment. Moreover, the existence of the threshold frequency in every case of photoelectric emission cannot be understood based on classical theory.

As has been mentioned at the beginning of this section, Einstein, in 1905 showed that the phenomenon of photoelectric emission could be understood in terms of Planck's quantum theory and provided a direct verification of the theory. He pointed out that the emission or absorption of energy takes place in quanta or photons of energy hf . When a photon of frequency f strikes the metal surface, it gives up its entire energy to the electrons on the surface of the metal. A portion of the energy is utilized in liberating the electron from the surface. This energy is called the work function and the frequency corresponding to this work function is called the threshold frequency, f_0 . The remaining energy will be given as kinetic energy to the emitted electron. Hence, we have from the conservation of energy:

$$\begin{aligned} &\text{Energy of a photon falling on the metal surface} \\ &= \text{Work function of the metal} \quad + \quad \text{Kinetic energy of the emitted electron} \end{aligned}$$

or

$$hf = \Phi + \frac{1}{2}mv_m^2, \quad (1.17)$$

where m is the mass of an electron and v_m is its speed; $\Phi = hf_0$ is the work function of the metal. This equation is known as the Einstein equation for photoelectric emission.

Combining Eqs. (1.16) and (1.17), we have

$$hf = hf_0 + eV_0, \quad (1.18)$$

which can be written as

$$f = f_0 + \left(\frac{e}{h}\right) V_0. \quad (1.19)$$

The three experimental results of photoelectricity can now be easily explained. It is evident from Eq. (1.18) that if the energy of the incident photon is less than the work function ($\Phi = hf_0$), no emission of electrons will take place whatever be the intensity of the incident radiation. Equation (1.19) indicates that if the frequency of the incident radiation (f) be plotted against the applied stopping potential V_0 , a straight line will be obtained and its intercept gives the value of the threshold frequency (f_0). The values of the slopes (e/h) corresponding to different metallic surface are calculated using Eq. (1.19) and found to be the same for all the cases. From the value of the slope, we get the Planck's constant as $6.55 \times 10^{-34} Js$, in agreement with the value obtained from the blackbody measurements. Further, Eqs. (1.17) and (1.18) show that the maximum kinetic energy is independent of the intensity of the incident radiation but varies directly with its frequency.

1.3 The Compton Effect

We have seen the Einstein's explanation for the photoelectric effect which implies that a quantum or a photon of light carries a definite amount of energy. The Compton Effect provides an example of a process in which a quantum of radiation carries energy as well as momentum. This gives the most convincing proof in support of the particle properties of radiation. In 1923, Compton investigated the scattering of X-rays by crystals and found that the wavelength of the scattered radiation is greater than the wavelength of the incident radiation. The observed change in wavelength of the scattered radiation is found to be independent of the wavelength of the incident radiation. This change in wavelength or frequency of the scattered radiation is known as the Compton Effect.

The classical electromagnetic theory when applied to this phenomenon showed that (a) the scattered X-rays should have the same wavelength as the incident one, (b) the scattered radiation should be symmetrical as regards the distribution of intensity. However, these classical concepts failed to explain the experimental findings of Compton.

Compton, however, explained the scattering phenomenon in the light of Planck's quantum theory by taking into consideration the particle characteristics of radiation. When an incident photon strikes an electron at rest, there will be an inelastic collision between the photon and the electron of the scatterer. The photon will transfer kinetic energy and momentum to the electron and therefore the scattered photon will have lower energy than the incident photon.

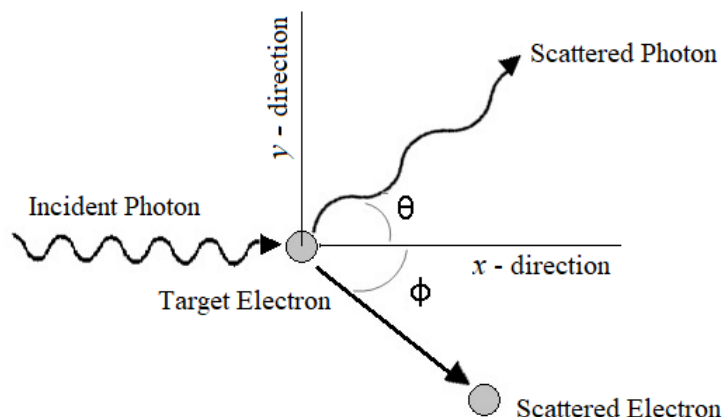


Fig. 1.5: Elastic collision of a photon with a free electron initially at rest.

Let a photon of energy hf and momentum hf/c collides with an electron of rest mass m_0 . Then, after collision, the scattered photon will have energy hf' and momentum hf'/c (Fig. 1.5). As the electron is

assumed to be at rest before collision, according to the theory of relativity its initial energy is m_0c^2 , while its momentum is zero. After the collision, if the velocity of the recoiling electron is v , then by the Einstein's theory of relativity, the mass of the electron is

$$m = \frac{m_0}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}}. \quad (1.20)$$

And the momentum of the scattered electron is

$$mv = \frac{m_0v}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}}. \quad (1.21)$$

Let θ and φ be the angles of the scattered photon and the scattered electron respectively with the direction of the incident photon. Then, by applying the principle of conservation of energy, we get

$$hf + m_0c^2 = hf' + mc^2,$$

which can be written as

$$h(f - f') = mc^2 - m_0c^2. \quad (1.22)$$

For the conservation of linear momentum, we get

$$\frac{hf}{c} + 0 = \frac{hf'}{c} \cos \theta + \frac{m_0v}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}} \cos \varphi, \quad (1.23)$$

for the x -component of the linear momentum and

$$0 + 0 = \frac{hf'}{c} \sin \theta - \frac{m_0v}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}} \sin \varphi. \quad (1.24)$$

for the y -component of the linear momentum.

Now, Eqs. (1.21) -(1.24) can be used to determine the change in wavelength. At first, unknown recoil angle φ is eliminated by squaring and adding Eqs. (1.23) and (1.24). Thus, we obtain

$$\frac{h^2}{c^2} (f'^2 + f^2 - 2f'f \cos \theta) = \frac{m_0^2 v^2}{\left(1 - \frac{v^2}{c^2}\right)}. \quad (1.25)$$

Squaring Eq. (1.22), we get

$$(h(f - f') + m_0c^2)^2 = m^2c^4,$$

or

$$h^2(f'^2 + f^2 - 2f'f) + 2m_0c^2h(f - f') + m_0^2c^4 = m^2c^4,$$

or

$$\begin{aligned}
 \frac{h^2}{c^2}(f'^2 + f^2 - 2ff') + 2m_0h(f - f') &= m^2c^2 - m_0^2c^2 \\
 &= \frac{m_0^2c^2}{\left(1 - \frac{v^2}{c^2}\right)} - m_0^2c^2 \\
 &= \frac{m_0^2v^2}{\left(1 - \frac{v^2}{c^2}\right)}. \tag{1.26}
 \end{aligned}$$

Subtracting Eq. (1.25) from Eq. (1.26) and dividing by $2m_0h$, we get

$$f - f' = \frac{hff'}{m_0c^2}(1 - \cos \theta).$$

Dividing both sides of the above equation by ff'/c , the above equation can be written as

$$\frac{c}{f'} - \frac{c}{f} = \frac{h}{m_0c}(1 - \cos \theta).$$

Now, using the formula $\lambda = c/f$, the above equation takes the following form for the change in wavelength due to the scattering of the photon with the electron:

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0c}(1 - \cos \theta). \tag{1.27}$$

Thus, the scattered wavelength is always greater than the incident wavelength. Equation (1.27) also shows that the change in wavelength is independent of the wavelength of the incident radiation as well as that of the scatterer. Compton was also able to show that the distribution in intensity of the scattered radiation was not symmetrical, all the scattering being in the forward direction, the backward scattering being practically nil. This asymmetry increases with the frequency of the incident radiation. Thus, Compton theory is in complete agreement with the experimental observation and provides convincing evidence in support of the particle nature of the radiation.

1.4 De Broglie Hypothesis & Wave-Particle Duality

De Broglie's Hypothesis

In the study of electromagnetic radiation, two different theories are alternatively in use; one based on the particle model and the other one on the wave model. The photoelectric effect and the Compton Effect can be explained by assuming the particle (corpuscular) nature of radiation, whereas reflection, refraction and diffraction can be explained on the basis of wave nature of radiation. It was de Broglie who in 1924 combined these two conflicting theories of radiation in his revolutionary hypothesis.

In support of the dual character of radiation, de Broglie suggested that every material particle is also associated with wave-like properties. He argued that if light can sometimes behave like particles, then it should be possible for matter to exhibit wave-like behavior under suitable conditions. This was verified experimentally by Germer, Davisson, and Thompson in 1927, during the study of electron diffraction. On the basis of these considerations, de Broglie establishes a relation between the momentum and the wavelength of a particle in motion. The energy E of a particle is treated in the same way as the energy of a photon, and is given by

$$E = hf, \quad (1.28)$$

where f is the frequency associated with the particle.

On the other hand, the energy E of a photon (particle like entity) of momentum p can be written as

$$E = pc, \quad (1.29)$$

where c is the speed of light. From these two equations, we get

$$hf = pc$$

which can be written as

$$\frac{hf}{c} = p. \quad (1.30)$$

Using $f\lambda = c$, we get $\lambda = c/f$, so Eq. (1.30) becomes

$$\lambda = \frac{h}{p}. \quad (1.31)$$

According to the wave and particle considered to be in the same footing, de Broglie proposed that wave nature also exists for any particle of mass m and velocity v , and its wavelength is given by the analogous relation ($p = mv$):

$$\lambda = \frac{h}{p} = \frac{h}{mv}. \quad (1.32)$$

This relation is known as the de Broglie equation and the wavelength λ associated with the particle is often called the ***de Broglie wavelength***.

The de Broglie equation, Eq. (1.32) shows that the fundamental relation between the momentum of the electron, considered as a particle, and the wavelength of the electron, considered as wave. From Eq. (1.32), it is obvious that $h/(mv)$ is negligible for relatively large objects (macro-particles) while for subatomic particles (microscopic particles) this quantity is not small enough to be considered negligible. So, the wave character is predominant in micro-particles whereas the particle character is more important in macro-particles.

The de Broglie concept does not imply that matter behaves like particles at one time and waves at another time. Rather, the wave nature has to be an inherent property of each particle. Micro-particles can exhibit particle or wavelike properties in some extent depending on the circumstances. In other words, the two distinct concepts of wave and particle of classical mechanics are inseparably mingled in micro-system, and they have a more general nature than their names indicate.

Determination of de Broglie Wavelength

Experimentally, de Broglie wavelength λ is determined as follows: when an electron is accelerated through a potential difference of V volts, its kinetic energy $mv^2/2$ will be equal to the electrostatic energy eV . Thus,

$$\frac{1}{2}mv^2 = eV. \quad (1.33)$$

From the above Eq. (1.33), we can write

$$mv = (2meV)^{1/2} \equiv p, \quad (1.34)$$

where m , v , and e are the mass, velocity, and charge of the electron respectively. Therefore, the de Broglie wavelength, given by Eq. (1.32), gives

$$\lambda = \frac{h}{(2meV)^{1/2}}. \quad (1.35)$$

Thus, from the Eq. (1.35), we can determine the de Broglie wavelength.

De Broglie Wave Phase Velocity

With what velocity do the de Broglie waves travel? Since we associate a de Broglie wave with a moving body, it is reasonable to expect that this wave travels at the same velocity v as that of the body. If we call the de Broglie wave velocity, the phase velocity v_{ph} , we may apply the usual formula

$$v_{ph} = f\lambda$$

to determine the value of v_{ph} , where f is the frequency. The wavelength λ is just the de Broglie wavelength

$$\lambda = \frac{h}{mv}.$$

We shall take the frequency f to be that specified by the quantum equation

$$E = hf.$$

Hence

$$f = \frac{E}{h}$$

or, since

$$E = mc^2$$

we have

$$f = \frac{mc^2}{h}.$$

The de Broglie wave velocity is therefore

$$\begin{aligned} v_{ph} &= f\lambda \\ &= \frac{mc^2}{h} \frac{h}{mv} \end{aligned}$$

or

$$v_{ph} = \frac{c^2}{v}. \quad (1.36)$$

Since the velocity v cannot be equal or exceed the velocity of light c , the de Broglie wave velocity v_{ph} is always greater than c ! Clearly v and v_{ph} are never equal for a moving body. To understand this unexpected result, we shall discuss next the wave velocity and group velocity. It is to be mentioned here that wave velocity is sometimes called phase velocity.

Wave Velocity and Group Velocity

The amplitude of the de Broglie waves that correspond to a moving body reflects the probability that it be found at a particular place at a particular time. The de Broglie waves cannot be represented simply by a sinusoidal wave

$$y(x, t) = A \cos(\omega t - kx),$$

which describes an infinite series of waves all with the same amplitude A . Instead, we expect the wave representation of a moving body to correspond to a *wave packet*, or *wave group*, like that shown in Fig. 1.6 whose constituent waves have amplitudes which vary with the likelihood of detecting the body.

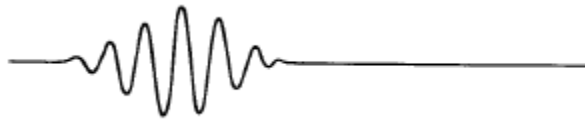


Fig. 1.6 A wave group.

[A familiar example of how wave groups come into being is the case of beats. When two sound waves of the same amplitude, but of slightly different frequencies, are produced simultaneously, the sound we hear has a frequency equal to the average of the two original frequencies and its amplitude rises and falls periodically. The amplitude fluctuations occur a number of times per second equal to the difference between the two original frequencies.]

A way of mathematically describing a wave group, then, is in terms of a series of individual waves, differing slightly in wavelength, whose interference with one another results in the variation on amplitude that defines the group shape. If the speeds of the waves are the same, the speed with which the wave group travels is identical with the common wave speed. However, if the wave speed varies with wavelength, the different individual waves do not proceed together, and the wave group has a speed different from that of the waves that compose it.

It is not difficult to compute the speed v_g with which a wave group travels. Let us suppose that a wave group arises from the combination of two waves with the same amplitude A but differing by an amount $d\omega$ in angular frequency and an amount dk in propagation constant. We may represent the original waves by the formulas

$$y_1 = A \cos(\omega t - kx),$$

$$y_2 = A \cos[(\omega + d\omega)t - (k + dk)x].$$

The resultant displacement y at any time t and any position x is the sum of y_1 and y_2 . Thus, we have the resultant displacement

$$y = y_1 + y_2 = 2A \cos \left[\frac{1}{2} (2\omega + d\omega)t - \frac{1}{2} (2k + dk)x \right] \cos \left[\frac{d\omega}{2} t - \frac{dk}{2} x \right].$$

Since $d\omega$ and dk are very small compared with ω and k respectively, we have

$$\begin{aligned} 2\omega + d\omega &\approx 2\omega, \\ 2k + dk &\approx 2k \end{aligned}$$

and

$$y = 2A \cos(\omega t - kx) \cos \left[\frac{d\omega}{2} t - \frac{dk}{2} x \right]. \quad (1.37)$$

Equation (1.37) represents a wave of angular frequency ω and propagation constant k that has superimposed upon it a modulation of angular frequency $d\omega/2$ and propagation constant $dk/2$. The effect of the modulation is to produce successive wave groups, as in Fig. 1.7.

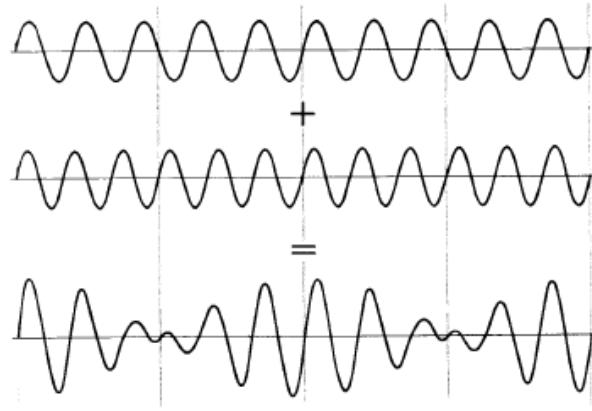


Fig. 1.7 The production of beats.

The wave velocity v_{ph} is

$$v_{ph} = \frac{\omega}{k} \quad (1.38)$$

while the velocity v_g of the wave group is

$$v_g = \frac{d\omega}{dk}. \quad (1.39)$$

In general, depending upon the way wave velocity varies with wavelength in a particular medium, the group velocity may be greater than or less than the wave velocity. If the wave velocity v_{ph} is the same for all wavelengths, the group velocity and wave velocity are the same.

Relation between Phase Velocity and Group Velocity

We know, the phase velocity is

$$v_{ph} = \frac{\omega}{k} \quad \Rightarrow \quad \omega = v_{ph}k,$$

Thus, the group velocity is

$$\begin{aligned}
 v_g &= \frac{d\omega}{dk} \\
 &= \frac{d}{dk}(v_{ph}k) \\
 &= v_{ph} + k \frac{dv_{ph}}{dk} \\
 &= v_{ph} + k \frac{dv_{ph}}{d\lambda} \frac{d\lambda}{dk} \\
 &= v_{ph} + k \frac{dv_{ph}}{d\lambda} \frac{d}{dk} \left(\frac{2\pi}{k} \right) \\
 &= v_{ph} + k \frac{dv_{ph}}{d\lambda} \left(-\frac{2\pi}{k^2} \right) \\
 &= v_{ph} - \lambda \frac{dv_{ph}}{d\lambda}, \\
 \Rightarrow \quad v_g &= v_{ph} - \lambda \frac{dv_{ph}}{d\lambda}. \tag{1.40}
 \end{aligned}$$

De Broglie Wave Group Travels with the same Velocity as the Body

From Eq. (1.39), the group velocity of the de Broglie wave is

$$v_g = \frac{d\omega}{dk},$$

where ω and k are respectively the angular frequency and the wave number respectively.

We know for a virtual particle with angular frequency ω , the energy E is given by

$$E = \hbar\omega. \quad [\text{where } \hbar = h/2\pi] \quad (1.41)$$

We also have from the de Broglie wavelength

$$\lambda = \frac{h}{p},$$

which can be written as

$$p = \frac{h}{\lambda} = \frac{2\pi\hbar}{\lambda} = \hbar \left(\frac{2\pi}{\lambda} \right),$$

$$\Rightarrow \quad p = \hbar k. \quad (1.42)$$

Thus, from Eqs. (1.41) and (1.42), we have

$$\frac{dE}{dp} = \frac{d(\hbar\omega)}{d(\hbar k)} = \frac{d\omega}{dk}.$$

Thus, the group velocity of the de Broglie wave can be written as

$$v_g = \frac{d\omega}{dk} = \frac{dE}{dp}. \quad (1.43)$$

For Non-Relativistic Particles

For non-relativistic particles, the total energy is

$$E = \frac{p^2}{2m} \quad \Rightarrow \quad \frac{dE}{dp} = \frac{p}{m} = \frac{mv}{m} = v. \quad (1.44)$$

Therefore,

$$v_g = \frac{d\omega}{dk} = \frac{dE}{dp} = v,$$

That is

$$v_g = v.$$

For Relativistic Particles

On the other hand, for relativistic particles

$$E^2 = p^2 c^2 + m_0^2 c^4. \quad (1.45)$$

Therefore,

$$\frac{dE}{dp} = \frac{pc^2}{E}. \quad (1.46)$$

But the energy E of the particle of mass m can also be written as

$$E = mc^2. \quad (1.47)$$

Also, the momentum p is

$$p = mv. \quad (1.48)$$

Therefore,

$$v_g = \frac{dE}{dp} = \frac{pc^2}{E} = \frac{mvc^2}{mc^2} = v,$$

that is

$$v_g = v.$$

Thus, we see that the de Broglie wave group associated with a moving body travel with the same velocity as the body.

1.5 Some Relevant Problems

Problem 1.1

What is the energy of the photon having a wavelength $1.5A$, given that $h = 6.627 \times 10^{-34} \text{ Js}$?

Solution

We know that the energy of a photon is

$$\begin{aligned} E &= hf \\ &= h \frac{c}{\lambda} \\ &= \frac{6.627 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{1.5 \times 10^{-10} \text{ m}} \\ &= 13.254 \times 10^{-16} \text{ J}. \end{aligned}$$

Problem 1.2

Calculate the average energy of an oscillator with frequency 10^{14} Hz at 1000 K . (Boltzmann constant $k_B = 1.38 \times 10^{-23} \text{ JK}^{-1}$).

Solution

We know that the average energy of an oscillator is

$$\langle E \rangle = \frac{hf}{e^{hf/k_B T} - 1}.$$

Here

$$\frac{hf}{k_B T} = \frac{(6.627 \times 10^{-34} \text{ Js}) \times (10^{14} \text{ s}^{-1})}{(1.38 \times 10^{-23} \text{ JK}^{-1}) \times (1000 \text{ K})} = 4.8.$$

Thus,

$$\begin{aligned} \langle E \rangle &= \frac{(6.627 \times 10^{-34} \text{ Js}) \times (10^{14} \text{ s}^{-1})}{(e^{4.8} - 1)} \\ &= 5.5 \times 10^{-22} \text{ J}. \end{aligned}$$

Problem 1.3

A beam of electrons is used to bombard gaseous hydrogen atoms. What is the minimum energy the electrons must have if the first member of the Balmer series, corresponding to a transition from $n = 3$ to $n = 2$ state, is to be emitted? ($4\pi\epsilon_0 = 1.113 \times 10^{-10} \text{ C}^2 \text{ J}^{-1} \text{ m}^{-1}$).

Solution

We know from the Bohr's theory that the energy of transition is

$$\Delta E = E_2 - E_1 = -\frac{2\pi^2 m e^4}{(4\pi\epsilon_0)^2 h^2} \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

Here $n_2 = 2$, $n_1 = 3$. Thus,

$$\begin{aligned} \Delta E &= -\frac{2\pi^2 m e^4}{(4\pi\epsilon_0)^2 h^2} \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right) \\ &= -\frac{2(3.14)^2 \times (9.1 \times 10^{-31} \text{ kg}) (1.6 \times 10^{-19} \text{ C})^4 0.22 \times 10^{-10} \text{ m}}{(1.113 \times 10^{-10} \text{ C}^2 \text{ J}^{-1} \text{ m}^{-1})^2 \times (6.627 \times 10^{-34} \text{ Js})^2} \left(\frac{1}{3^2} - \frac{1}{2^2} \right) \\ &= 2.98 \times 10^{-19} \text{ J}. \end{aligned}$$

Problem 1.4

The threshold wavelength for photoelectric emission in tungsten is 2300Å. What wavelength of light must be used for electrons with a maximum energy of 1.5eV to be ejected?

Solution

From equation

$$\frac{1}{2}mv^2 + \Phi = hf$$

or

$$T_{max} + \Phi = hf,$$

where T_{max} is the maximum photoelectron energy

$$T_{max} = \frac{1}{2}mv^2.$$

Since $f = c/\lambda$, we have

$$h\frac{c}{\lambda} = T_{max} + \Phi.$$

After substituting the known values of the parameters, we have

$$\begin{aligned} \frac{6.63 \times 10^{-34} \text{J.s} \times 3 \times 10^8 \text{m.s}^{-1}}{\lambda} \\ = 1.5 \text{eV} \times 1.6 \times 10^{-19} \text{J/eV} + \frac{6.63 \times 10^{-34} \text{J.s} \times 3 \times 10^8 \text{m.s}^{-1}}{2300 \text{\AA} \times 10^{-10} \text{m/\AA}}, \end{aligned}$$

or

$$\frac{19.89 \times 10^{-26} \text{J.m}}{\lambda} = 0.240 \times 10^{-18} \text{J} + 0.865 \times 10^{-18} \text{J} = 1.11 \times 10^{-18} \text{J},$$

from which we obtain

$$\lambda = \frac{19.89 \times 10^{-26} \text{J.m}}{1.11 \times 10^{-18} \text{J}} = 1.8 \times 10^{-7} \text{m} \equiv 1800 \text{\AA}.$$

Problem 1.5

The work function of potassium is 2.0eV. When ultraviolet light of wavelength 3,500Å (1Å = 1 Angstrom unit = 10^{-10} m) falls on a potassium surface, what is the maximum energy in eV of the photoelectrons?

Solution

From equation

$$\frac{1}{2}mv_m^2 + \Phi = T_{max}$$

where T_{max} is the maximum photoelectron energy, $T_{max} = mv_m^2/2$. Thus, we have

$$hf = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \text{J.s} \times 3 \times 10^8 \text{m.s}^{-1} \times 10^{10} \text{\AA m}^{-1}}{3500 \text{\AA}} = 5.7 \times 10^{-19} \text{J}.$$

To convert this energy from joules to electron volts, we recall that

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{J}$$

and so

$$hf = \frac{5.7 \times 10^{-19} \text{J}}{1.6 \times 10^{-19} \text{J/eV}} = 3.6 \text{ eV}.$$

Hence the maximum photoelectron energy is

$$T_{max} = hf - \Phi = 3.6 \text{ eV} - 2.0 \text{ eV} = 1.6 \text{ eV}.$$

Problem 1.6

The threshold wavelength for photoelectric emission in tungsten is 2300Å. What wavelength of light must be used to eject electrons with a maximum energy of 1.5eV? (Planck's constant, $h = 6.627 \times 10^{-34} \text{ Js}$)

Solution

The kinetic energy is

$$E = hf - hf_0 = h(f - f_0) = h\left(\frac{c}{\lambda} - \frac{c}{\lambda_0}\right) = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right) \quad [\text{work function, } \Phi = hf_0 = hc/\lambda_0].$$

or

$$\frac{1}{\lambda} = \frac{E}{hf} + \frac{1}{\lambda_0},$$

where

$$E = 1.5 \text{ eV} = 1.5 \times 1.6 \times 10^{-19} \text{ J}$$

$$\lambda_0 = 2300 \text{ Å} = 23 \times 10^{-8} \text{ m}.$$

Thus,

$$\frac{1}{\lambda} = \frac{(1.5 \times 1.6 \times 10^{-19} \text{ J})}{(6.627 \times 10^{-34} \text{ Js}) (3 \times 10^8 \text{ ms}^{-1})} + \frac{1}{23 \times 10^{-8} \text{ m}} = 0.5508 \times 10^7 \text{ m}^{-1}.$$

Therefore,

$$\lambda = 1.8155 \times 10^{-7} \text{ m}.$$

Problem 1.7

What is the frequency of an X-ray photon whose momentum is $1.1 \times 10^{-23} \text{ kg.m.s}^{-1}$?

Solution

We know the relation

$$E = pc = hf,$$

from which we have

$$f = \frac{pc}{h} = \frac{1.1 \times 10^{-23} \text{ kg.m.s}^{-1} \times 3 \times 10^8 \text{ m.s}^{-1}}{6.63 \times 10^{-34} \text{ J.s}} = 4.98 \times 10^{18} \text{ s}^{-1}.$$

Problem 1.8

An X-ray photon of initial frequency $3 \times 10^{19} \text{ s}^{-1}$ collides with an electron and is scattered through 90° . Find the frequency of the photon after collision.

Solution

We know the formula

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos \theta).$$

We have

$$\frac{c}{f'} - \frac{c}{f} = \frac{h}{m_0 c} (1 - \cos \theta), \quad (c = \nu\lambda)$$

or

$$\frac{1}{f'} - \frac{1}{f} = \frac{h}{m_0 c^2} (1 - \cos \theta),$$

or

$$\begin{aligned}
\frac{1}{f'} &= \frac{h}{m_0 c^2} (1 - \cos \theta) + \frac{1}{f} \\
&= \frac{6.63 \times 10^{-34} \text{ J}\cdot\text{s}}{9.1 \times 10^{-31} \text{ kg} \times (3.0 \times 10^8 \text{ m}\cdot\text{s}^{-1})^2} [1 - \cos(90^\circ)] + \frac{1}{3 \times 10^{19} \text{ s}^{-1}} \\
&= 0.08 \times 10^{-19} \text{ s} + 0.33 \times 10^{-19} \text{ s} \\
&= 0.41 \times 10^{-19} \text{ s}.
\end{aligned}$$

Thus, the frequency of the photon after collision is
 $f' = 2.44 \times 10^{19} \text{ s}^{-1}$.

Problem 1.9

A beam of X-rays is scattered by loosely bound electrons at 45° to the direction of the beam. The wavelength of the scattered X-rays is $0.22 \text{ \AA}$. Find the wavelength of the incident X-ray beam. (The Planck's constant $h = 6.627 \times 10^{-34} \text{ J}\cdot\text{s}$).

Solution

We know

$$\lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos \theta).$$

Here

$$\lambda' = 0.22 \text{ \AA} = 0.22 \times 10^{-10} \text{ m}, \quad \theta = 45^\circ.$$

Thus

$$\begin{aligned}
\lambda &= \lambda' - \frac{h}{m_0 c} (1 - \cos \theta) \\
&= 0.22 \times 10^{-10} \text{ m} - \frac{(6.627 \times 10^{-34} \text{ J}\cdot\text{s})}{(9.1 \times 10^{-31} \text{ kg}) \times (3 \times 10^8 \text{ m}\cdot\text{s}^{-1})} (1 - \cos 45^\circ) \\
&= 0.2129 \times 10^{-10} \text{ m}.
\end{aligned}$$

Problem 1.10

An electron is characterized by applying a potential difference of 1000 V . What is the de Broglie wavelength associated with it? (The Planck's constant $h = 6.627 \times 10^{-34} \text{ J}\cdot\text{s}$).

Solution

We know the de Broglie wavelength is

$$\lambda = \frac{h}{(2meV)^{1/2}}.$$

Here

$$\begin{aligned}
\lambda &= \frac{h}{(2meV)^{1/2}} \\
&= \frac{6.627 \times 10^{-34} \text{ J}\cdot\text{s}}{(2 \times 9.1 \times 10^{-31} \text{ kg} \times 1.6 \times 10^{-19} \text{ C} \times 1000 \text{ V})^{1/2}} \\
&= 38.7 \times 10^{-12} \text{ m}.
\end{aligned}$$

Problem 1.11

An electron has kinetic energy 54 eV. Find the de Broglie wavelength of the electron.

Solution

We know the expression for kinetic energy of the electron is

$$T = \frac{1}{2}mv^2.$$

Thus the momentum of the electron is

$$\begin{aligned} mv &= \sqrt{2mT} \\ &= \sqrt{2 \times 9.1 \times 10^{-31} \text{kg} \times 54 \text{ eV} \times 1.6 \times 10^{-19} \text{J/eV}} \\ &= 4.0 \times 10^{-24} \text{kg.m.s}^{-1}. \end{aligned}$$

The electron wavelength is therefore

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34} \text{J.s}}{4.0 \times 10^{-24} \text{kg.m.s}^{-1}} = 1.66 \times 10^{-10} \text{m} = 1.66 \text{\AA}.$$

Problem 1.12

Calculate the de Broglie wavelength associated with an electron in the first Bohr orbit. (The Planck's constant $h = 6.627 \times 10^{-34} \text{ J.s}$ and $4\pi\epsilon_0 = 1.113 \times 10^{-10} \text{ C}^2\text{J}^{-1}\text{m}^{-1}$).

Solution

We know

$$mvr = \frac{nh}{2\pi}, \quad \text{and} \quad \frac{Ze^2}{4\pi\epsilon_0 r^2} = \frac{mv^2}{r}.$$

Equating these two relations, we get

$$v = \frac{e^2}{4\pi\epsilon_0} \frac{2\pi}{nh} = \frac{(1.6 \times 10^{-19} \text{ C})^2}{1.113 \times 10^{-10} \text{ C}^2\text{J}^{-1}\text{m}^{-1}} \frac{2 \times 3.14}{1 \times 6.627 \times 10^{-34} \text{ J.s}} = 2.19 \times 10^6 \text{ ms}^{-1}.$$

Thus, the wavelength is

$$\lambda = \frac{h}{mv} = \frac{6.627 \times 10^{-34} \text{ J.s}}{9.1 \times 10^{-31} \text{ kg} \times 2.19 \times 10^6 \text{ ms}^{-1}} = 3.32 \times 10^{-10} \text{ m}.$$

Problem 1.13

Find the de Broglie wavelength of a 1 kg object whose speed is 1 m/s.

Solution

We know the expression for the de Broglie wavelength is

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34} \text{J.s}}{1 \text{ kg} \times 1 \text{ m.s}^{-1}} = 6.63 \times 10^{-34} \text{m}.$$

Problem 1.14

Discuss the quantum theory of radiation. Write down the expressions for the energy and momentum of photons. Find the energy of a photon with wavelength 7500Å.

Problem 1.15

Write down the Planck's hypothesis. Develop the Planck's radiation formula.

Problem 1.16

Write down the Planck's radiation formula in terms of frequency. From this, express the formula in terms of wavelength. Also show that when the wavelength approaches infinity, the Planck's law reduces to the Rayleigh-Jeans law.

Problem 1.17

What is photoelectric effect? Discuss the experimental findings of the photoelectric effect.

Problem 1.18

The work function of potassium is 2eV. If light of wavelength 3000Å falls on a potassium plate then find the maximum kinetic energy of the electrons emitted from it.

Problem 1.19

What is Compton Effect? Write down the conservation equations pertaining to Compton Effect.

Problem 1.20

Derive an expression for the change in wavelength of a photon when it strikes an electron in the case of Compton Effect.

Problem 1.21

An x-ray with wavelength 0.21Å strikes an electron and is scattered at an angle of 45° . Find the wavelength of the scattered x-ray.

Problem 1.22

Explain the wave properties of particles. Find the de Broglie wavelength for a particle of mass 1.06×10^{-27} kg with kinetic energy 12keV.

Problem 1.23

- What do you mean by the de Broglie wavelength? Discuss.
- At 300K temperature the kinetic energy of a neutron is 0.04eV. Find the corresponding de Broglie wavelength. (Mass of a neutron, $m_n = 1.675 \times 10^{-27}$ kg)

Problem 1.24

- Discuss the wave-particle duality.
- Find a relation between the group velocity of the de Broglie wave and the corresponding particle velocity.
- Find the de Broglie wavelength of an electron with energy 50eV.

Problem 1.25

The de Broglie wavelength of a hydrogen atom is 1.105Å. Find the corresponding kinetic energy of the atom. (Mass of a proton, $m_p = 1.673 \times 10^{-27}$ kg)

Problem 1.26

Find the relation between the phase velocity and group velocity.
